Rajagopal Somaskandan

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Summary

M.Sc. Computer Science student with practical experience in software development, data processing, and machine learning through internships and projects. Strong foundation in programming, SQL, object-oriented programming, data structures, databases, and REST APIs, with experience applying these skills to build data-driven and software-based solutions.

Education

Sathyabama Institute of Science and Technology 2027
M.Sc. Computer Science *Chennai, India*

Sathyabama Institute of Science and Technology 2025
B.Sc. Computer Science with AI – ***First Class*** *Chennai, India*

Technical Skills

Languages: Python, SQL
Backend & Databases: FastAPI, REST APIs, PostgreSQL, MongoDB
Data & Machine Learning: NumPy, Pandas, Scikit-learn, XGBoost, LightGBM, CatBoost, SHAP
AI & Deep Learning: PyTorch, Transformers, Sentence-Transformers, RAG, FAISS, Computer Vision
Tools: Git, GitHub, Docker, Linux, Jupyter, Selenium, COLMAP, FFmpeg

Projects

SourceUp: Supplier Search & Ranking System Research Project – Submitted for IEEE

- Developed a data-driven **supplier search and recommendation application** for **1.27M supplier records**, combining semantic search, ranking, and explainability.
- Built a retrieval and ranking pipeline using **Python, SBERT, FAISS, and LightGBM LambdaRank** with **23K+ supplier–query pairs**.
- Implemented evaluation and **SHAP-based explanations** to make supplier recommendations more interpretable.

FinGuard Pro: Transaction Fraud Detection System Published, IEEE ICAAIC 2025

- Developed a **Python-based fraud detection pipeline** for **100K financial transactions**, including data preprocessing, model training, evaluation, and explainability.
- Compared **XGBoost, LightGBM, and CatBoost**, achieving **94% recall** and **0.9951 ROC-AUC**.
- Integrated **SHAP** to provide feature-level explanations for model predictions.

S.A.F.E: Crowd Anomaly Detection System Design Thinking Project, 2026

- Designed a **real-time anomaly detection application** using benchmark crowd datasets and evaluated multiple detection approaches to improve cross-dataset generalization and reduce false positives.
- Selected **Median Absolute Deviation (MAD)** based on cross-dataset evaluation, achieving **0.8391 ROC-AUC** and reducing false positives by **40–60%**.
- Developed a **FastAPI backend** for serving anomaly detection results and explainable alerts.

Experience

Production AI Developer (Intern) Mar 2026 – Jun 2026
InnoXR Labs Pvt. Ltd. *Chennai, India*

- Developed an end-to-end **Python ML pipeline** for single-video **3D reconstruction**, integrating video preprocessing, **COLMAP**, Gaussian Splatting, and GPU-based execution to automate the reconstruction workflow.
- Built **FastAPI backend services** and integrated a **Three.js** viewer for interactive 3D visualization, enabling end-to-end evaluation of reconstructed scenes using Tanks and Temples data.

Data Analyst Intern Jul 2024 – Oct 2024
India Advocacy *Chennai, India*

- Developed **Java and Selenium ETL pipelines** for processing more than **1M records**, including data extraction, cleaning, transformation, and preprocessing.

Achievements & Hackathons

Best Industry Innovation Award, Industry Innovation '26 – National-level hackathon, Sathyabama Institute of Science and Technology, competing against **63+ teams** (Mar 2026)

4th Place, Origin 24-Hour Hackathon – SIMATS Engineering, Chennai (Apr 2026)

Certifications

AI & ML | *Space Zee Technologies* Jul 2024
Python Core | *Space Zee Technologies* Feb 2024